

I had an entire career in the Air Force managing flight records, runway activities, and the like and while accurate, this diagram still feels confusing. Maybe this diagram is poorly laid out or maybe aviation remains pretty complicated XD

1a -

Relationships

Departure Airport and Arrival Airport

1 Airport : N Flight Legs

Departs and Arrives

1 Airport : N Leg Instance

Assigned

1 Airplane : N Leg Instance

Reservation

N Seats : 1 Leg Instance

Can Land

M Airports : N Airplane Types

1b -

Weak Entities

Flight Leg

Entity: Flight

Relationship: Leg

Leg Instance

Entity: Flight Leg

Relationship: Instance Of

Seat

Entity: Leg Instance

Relationship: Reservation

Fare

Entity: Flight

Relationship: Fares

2a -

Designing a database system for a soccer club, first we would need to consider the club's size. How many people are coming to watch the soccer club? Is this a premier league team of the size that you would see in Ted Lasso or the local K-6 soccer league with the majority of the fans being the parents of the players?

The club would know the rough size and number of roles that exist within the organization already. If it is a larger club, it would need roles assigned to management, coach, waterboys, janitors, sports physiologist and psychiatrist, various administrative roles, and player roles. For a smaller club, the roles of management, snack mom, coach, players, and maybe a few other ancillary roles would suffice.

Most of these roles would report to various layers of the administration, such as the players reporting to the coaches, the support staff reporting to the management, and the players buying tickets that marketing would advertise.

2b - &c

